

React:

- React is a front to JavaScript library for building UI, especially SPA, using reusable components.
- Used to build dynamic and interactive web interface.
- works using components.

Features:

- 1) Component based architecture.
- a) Virtual DOM
- 3) JSX.
- 4) Reusable component.

MongoDB

(command). (after installation run this command on CMD).

- show dbs // to show existing databases.
- use myDatabase // use already existing database.
- db.dropDatabase() // to drop current using database.

show collections

- db.createCollection ("collectionName")
- db.student.drop()

- db.student.insertOne ({ // Here student is a collection.
 name: "ABU",
 age: 22,
 course: "B.Tech" // insert only one data in collection.
 })

- db.student.insertMany ([// insert multiple data in collection.
 {
 },
 {, ' },
 { }])

MongoDB is nosql database that stores data in a flexible JSON like format (BSON) instead of traditional tables.

SQL	MongoDB
Table	Collection
Row	Document
Column	Field

```
db.student.find()
db.student.find({age: 22})
```

```
db.student.updateOne(
  { name: "ABC" },
  { $set: {age: 23} }
)
```

```
db.student.updateMany(
  { course: "B.Tech" },
  { $set: {status: "active"} }
)
```

```
db.student.deleteOne({ name: "ABC" })
db.student.deleteMany({ age: { $lt: 21 } })
```

lt = less than

Common \$ operators :

\$set : value update karna
\$inc : value increase karna
\$lt : less than
\$gt : greater than
\$push : array me add karna
\$unset : field karta dena.

Example

```
db.user.find(
  { age: { $gt: 18 } }
)
```

ETL (Extract, Transform, Load)

- Extract → data is taken from different sources
- Transform → data is cleaned / converted into proper format.
- Load → Data is stored in database.

Example: Customer order data → cleaned → stored in system

XML (Extensible Markup Language)

used to store and transfer data in a structured way.

Ex: < order >

< customer > Teemanna < / customer >
 < product > Laptop < / product >
 < price > 25000 < / price >

< / order

It stores data using tags (fields + values).

Rules of XML

- Must have one root element
- Proper opening and closing tags.
- ⇒ Case sensitive
- Proper nesting.

XML

stores data
 custom tags
 focus on structure

HTML

displays data
 fixed tags.
 focus on design

DTD (Document Type Definition) (Kind of schema).

- It defines rules / structure of XML
- Checks XML is valid or not.

Ex: DTD ensures <Order> must contain <customer>, <products> etc.

Why validation?

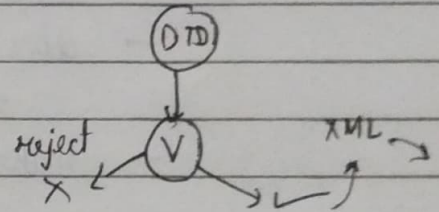
- Ensures correctness & consistency.

Where is XML used?

- Web services (APIs)
- Android layouts.
- Configuration files.
- Banking systems.
- Data exchange between systems.

Types of Data:

- Structured → Tables (DB)
- Unstructured → Text, images
- Semi structured → XML, JSON.



- XML : stores data
- DTD : defines rules for XML
- Validation : checks if XML follows rules.

Ques 10. How is XML data displayed on webpage?

XML → data fetch → JS → CSS → HTML.

XML → stores data

JS → extract, read data.

HTML → displays data on screen.

CSS → to beautify / style.

Ques Role of JS in XML

- used to read and process data
- converts XML ^{text} tags into readable code using parser
- extracts specific data
- get element by tag name.
- displays extracted data in web page.

Ques DOM parser :

- JS object used to convert XML texts into structured document.
- helps JS to understand XML texts.

Ques XSLT :

CSS

- used for styling
- does not change structure
- makes data look good.

XSLT

- used for transformation
- can change the structure
- converts XML to HTML

Ques How Apps use XML in real life?
Give one example.

Ques What is XSLT: Extensible stylesheet language Transformation

- used to convert XML data → HTML format

C/CD = $\rightarrow$ continuous Integration & continuous deployment.
 $\rightarrow$ automation pipeline.

JERRA: Project Management Software

1. Data Modelling Process of designing how data is structure stored and related in a system.

2. Rendering: Process of converting data or models into a visual or displayable form

1. Client side Rendering (CSR)

Browser renders webpage using JS.

2. Server side Rendering (SSR)

server generates full HTML page and sends it to browser.

3. Static Site Rendering

pages are pre-built at build time and served as static files, no rendering needed.

Feature	Client Side Rendering	Server SR	Static Site Rendering
Rendering speed	Browser slow - first	Server fast	pre-built fastest
ACD	low	high	high

Testing: functional, and unit testing

C/CD: code $\rightarrow$ build $\rightarrow$ test $\rightarrow$ deploy $\rightarrow$ live
 automation pipeline.